

Sum of Subspaces

The Sum of two subspaces U and W of a vector space V is defined as

$$U + W = \{u + w \mid u \in U, w \in W\}$$

Definition: Let U, W be subspaces of V . Then V is said to be the direct sum of U and W and we write $V = U \oplus W$

if $V = U + W$ and $U \cap W = \{0\}$.

Let U, W be subspaces of V . Then $V = U \oplus W$ if and only if for every $v \in V$ there exist unique vectors $u \in U$ and $w \in W$ such that $v = u + w$.

Properties :

1. The zero vector ' 0 ' of V is in $U + W$.
2. For any $u, w \in U + W$, we have $u + v \in U + W$.

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Example : Consider $U = \{ (a, 0, 0) / a \in \mathbb{R} \}$
 $W = \{ (0, b, c) / b, c \in \mathbb{R} \}$

Thus $V = U + W = \{ (a, b, c) / a, b, c \in \mathbb{R} \}$

Hence the direct sum of subspaces U and W results into vector space